

REGULATION OF GASTRIC SECRETION (METHODS OF STUDY)

- Regulation of gastric secretion and intestinal secretion has been studied by various **experimental animal procedures**.

EXPERIMENTAL METHODS OF STUDY

Experimental Model	Preparation & Anatomy	Nerve & Vascular Supply	Experimental Uses
1. Pavlov Pouch	- Designed by Russian scientist Ivan Pavlov (in a dog). - A small bag-like pouch made from a part of the stomach. Incompletely separated from the main portion of the stomach. - An isthmus connects the two parts (larger stomach and smaller pouch).	Parasympathetic supply: Intact (receives Vagus nerve fibers through the isthmus). Sympathetic supply: Intact (travels along blood vessels). Blood supply: Intact.	- Used to demonstrate different phases of gastric secretion. - Particularly useful in studying the Cephalic Phase to demonstrate the role of the Vagus nerve in gastric secretion.
2. Heidenhain Pouch	- Modified Pavlov pouch. Completely separated from the main portion of the stomach by cutting the isthmus without damaging blood vessels.	Parasympathetic supply: Absent (Vagal fibers severed). Sympathetic supply: Intact (remains through intact blood vessels). Blood supply: Intact.	- Used to demonstrate the role of sympathetic nerves. - Used to study the hormonal regulation of gastric secretion after vagotomy (cutting the vagus nerve).

Experimental Model	Preparation & Anatomy	Nerve & Vascular Supply	Experimental Uses
3. Bickel Pouch	• Prepared by cutting sympathetic nerve fibers along with removing surrounding blood vessels. • Results in a totally denervated pouch.	• Parasympathetic supply: Absent. • Sympathetic supply: Absent. • (<i>Totally denervated</i>).	• Used to demonstrate the pure role of hormones in gastric secretion without any neural influence.
4. Farrell and Ivy Pouch	• Prepared by completely removing the Bickel pouch from the stomach. • Autotransplanted into the subcutaneous tissue of the abdominal wall or thoracic wall in the same animal. • New local blood vessels develop over time to supply the transplanted pouch.	• Nerve supply: Completely absent (no neural connection to the gut). • Vascular supply: Revascularized by new local blood vessels.	• Used to study the pure hormonal mechanisms operating during the Gastric and Intestinal phases of gastric secretion.
5. Sham Feeding	• Experimental method involving false feeding. • The esophagus is transected and the cut ends are brought out to the exterior of the neck. • Food taken orally by the animal falls out through the upper esophageal fistula without reaching the stomach.	• Intact neural pathways (Vagus nerve active during chewing/tasting).	• Used to demonstrate the stimulation of gastric juice secretion during the Cephalic Phase. • <i>Evidence:</i> After vagotomy, sham feeding fails to induce gastric secretion, confirming the critical role of the Vagus nerve in the cephalic phase.

COMPARATIVE INNERVATION OVERVIEW

Gastric Pouch Models

Pavlov Pouch

- Vagus Nerve: INTACT
- Sympathetic: INTACT
- Blood Supply: INTACT

Heidenhain Pouch

- Vagus Nerve: CUT
- Sympathetic: INTACT
- Blood Supply: INTACT

Bickel Pouch

- Vagus Nerve: CUT
- Sympathetic: CUT
- (Totally Denervated)

Farrell & Ivy Pouch

- Subcutaneous Transplant
 - Pure Hormonal Assessment

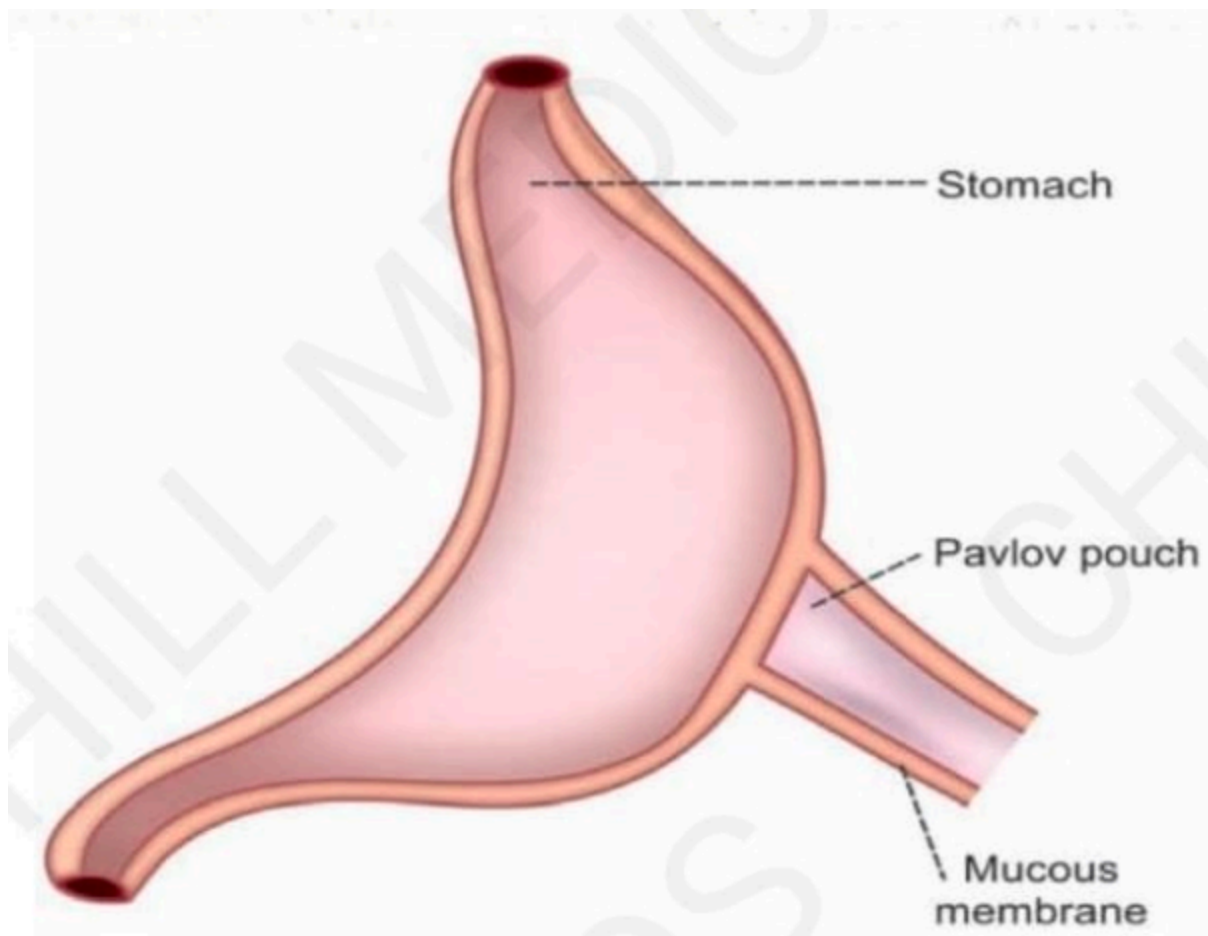


FIGURE 38.5: Pavlov pouch